

**Weil's theorem on the construction of a group from
a rational law**
SGA 3, Exposé XVIII

English draft

Michael Artin

About this draft

This is an unofficial English translation of Exposé XVIII, “Weil’s theorem on the construction of a group from a rational law” from *Schémas en groupes* (SGA 3). The source is the recomposition edited by P. Gille and P. Polo, PDF `Expo18.pdf` (18 pages), from <https://webusers.imj-prg.fr/~patrick.polo/SGA3/>. The seminar was directed by M. Demazure and A. Grothendieck (IHÉS, 1962–1964).

Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a `% typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked `% typo?:`. Scholarly proofreading remains outstanding.

The translator’s contribution is licensed under the MIT License. The French original remains copyright of its authors and publishers.

Contents

About this draft	iii
Exposé XVIII. Weil's theorem on the construction of a group from a rational law	1
0. Introduction	1
1. "Recollections" on rational maps	1
2. Local determination of a morphism of groups	3
3. Construction of a group from a rational law	5
Bibliography	14

Exposé XVIII. Weil's theorem on the construction of a group from a rational law

by Michael Artin

0. Introduction

This exposé is devoted to the well-known theorem of Weil [1] which gives the construction of a group variety from a birational law. It seems that the generalization of this result to the case where the base is the spectrum of a discrete valuation ring was already known to several people; one may, for example, see [2]. Here we shall prove the theorem for a flat group of finite presentation over an arbitrary base prescheme S . Since one must make the hypotheses with a little more care, the statement is not in the form given by Weil; but when S is the spectrum of a field, one sees that the form given here is essentially equivalent to Weil's. We use the following suggestion of Grothendieck: given a “group germ” X over a prescheme S , one must construct the group as the quotient of $X \times_S X$ by a suitable equivalence relation.

1. “Recollections” on rational maps

Let X/S be a relative prescheme and $U \subseteq X$ an open subset. One says that U is *schematically dense in X relative to S* , or that $U \hookrightarrow X$ is *relatively schematically dense*, if for every base change $S' \rightarrow S$ the open subset $U \times_S S'$ is schematically dense in $X \times_S S'$. For the definition and properties of this notion one refers to Exp. IX, § 4. ^(*)

Proposition 1.1. (i) *A finite intersection, as well as the union of a nonempty family of open subsets schematically dense relative to S , is schematically dense relative to S .*

(ii) *If $U \subseteq X$ is schematically dense relative to S and $S \rightarrow T$ is a morphism, then $U_T \subseteq X_T$ is schematically dense relative to T .*

(iii) *Let $U \subseteq V \subseteq X$ be open immersions. For U to be schematically dense in X relative to S , it is necessary and sufficient that it be so in V and that V be so in X .*

(iv) *If $U \subseteq X$ and $V \subseteq Y$ are relatively schematically dense, then $U \times_S V$ is schematically dense in $X \times_S Y$ relative to S .*

In Exp. IX one finds the following criterion:

Proposition 1.2. *Let X/S be locally of finite presentation and flat, and let U be an open subset of X . If, for each $s \in S$, the fiber U_s is schematically dense in X_s , then U is schematically dense in X relative to S .*

In particular:

Corollary 1.3. *If X/S is locally of finite presentation and flat, and each fiber X_s is “without embedded components”, then U is schematically dense in X relative to S if and only if, for each $s \in S$, U_s is dense in X_s in the topological sense.*

⁽⁰⁾Version of 8 May 2009.

^(*)Cf. also EGA IV₃, 11.9 and 11.10 (in particular 11.10.8), where one says “universally schematically dense relative to S ”.

One easily deduces from the definition the

Proposition 1.4. *Let $U \subseteq X$ be schematically dense relative to S , and let $f, g : X \rightarrow Y$ be two morphisms of functors over S , where Y satisfies one of the following conditions:*

- (i) *Y is a prescheme over S , and each fiber Y_s is separated.*
- (ii) *Y is a presheaf⁽¹⁾ over S such that the diagonal morphism $Y \rightarrow Y \times_S Y$ is represented by a closed immersion.*⁽²⁾

Then, if $f = g$ on U , one has $f = g$.

PROOF. In case (i), the standard argument after IX 4.1 shows that f, g are equal on each fiber; hence the subprescheme $\text{Ker}(f, g)$ of X is set-theoretically equal to X , hence is closed, and since it contains U , it is equal to X , i.e. $f = g$. Case (ii) is proved as in *loc. cit.*⁽³⁾ $\square$

Definition 1.5.^(*) Let X be a prescheme over S , and Y a presheaf over S . One calls a rational map $f : X \rightarrow Y$ over S an equivalence class of morphisms $f : U \rightarrow Y$ over S , where U is an open subset of X schematically dense relative to S , and where one sets $(f : U \rightarrow Y) \sim (f' : U' \rightarrow Y)$ if and only if there exists an open subset $U'' \subseteq U' \cap U$, schematically dense relative to S , such that $f = f'$ on U'' .

This definition is made in such a way that one can define in an evident manner the rational map $f \times_S S'$ for every base extension $S' \rightarrow S$.

If U is an open subset of X , one says that the rational map f is *defined on U* if there exists a morphism representing f whose set of definition contains U .

Definition 1.5.1.⁽⁴⁾ If Y satisfies one of conditions (i), (ii) of 1.4, it is clear that there exists a largest open subset U of X such that f is defined on U , and this U is schematically dense relative to S . One calls it the *domain of definition of f over S* , and denotes it by $\text{Dom}(f)$.

This notion does not commute with base changes, but one has

Proposition 1.6. *Let X be an S -prescheme and Y an S -functor satisfying one of hypotheses (i), (ii) of 1.4. Let $f : X \rightarrow Y$ be a rational map over S , $S' \rightarrow S$ a flat morphism locally of finite presentation, and $f' = f \times_S S'$. Then*

$$\text{Dom}(f') = \text{Dom}(f) \times_S S'.$$

PROOF. Set $U = \text{Dom}(f)$. It is clear that $V' = \text{Dom}(f')$ contains $U \times_S S'$. Let V be the image of V' , which is an open subset of X , since $X' \rightarrow X$ is open. One must prove that $V = U$, that is, one must find a morphism $V \rightarrow Y$ representing f . Set $S'' = S' \times_S S'$,

$$X'' = X \times_S S'', \quad U'' = U \times_S S'', \quad V'' = V' \times_{V'} V'.$$

Then U'' is schematically dense in X'' relative to S'' , hence U'' is schematically dense in V'' relative to S , since $U'' \subseteq V'' \subseteq X''$. The restriction of $f' : V' \rightarrow Y'$ to U' is deduced from $f : U \rightarrow Y$ by base change. The two morphisms $V'' \rightarrow Y$ deduced from g by base change are equal on U'' , hence equal. Now, since $V' \rightarrow V$ is flat and locally of finite presentation, it is fppf-covering (Exp. IV 6.3), and one finds the morphism $V \rightarrow Y$ by descent. $\square$

⁽¹⁾**N.D.E.** Specify for which topology: a priori (fpqc).

⁽²⁾**N.D.E.** Let us recall here the definition of a closed immersion. A morphism of S -functors $F \rightarrow G$ is relatively representable if, for every S -morphism $T \rightarrow G$, where T is an S -prescheme, the T -functor $F_T = F \times_S T$ is represented by a T -prescheme. A morphism of S -functors $F \rightarrow G$ is an open (resp. closed) immersion if it is relatively representable and, for every S -morphism $T \rightarrow G$, where T is an S -prescheme, the T -morphism $F_T \rightarrow T$ is an open (resp. closed) immersion.

⁽³⁾**N.D.E.** Give details of this point...

^(*)Cf. EGA IV₄, 20.5, where one says "pseudo-morphism from X to Y relative to S ", so as not to conflict with EGA I, 7.12.

⁽⁴⁾**N.D.E.** The number 1.5.1 has been added.

We shall frequently use the following triviality:

Proposition 1.7. *Let X/S be faithfully flat, locally of finite presentation, and let $U \subseteq X$ be a relatively schematically dense open subset. Then there exists a base extension $S' \rightarrow S$ which is fppf-covering, and a section $x \in X(S')$ which is contained in $U(S')$.*

PROOF. Indeed, U/S is fppf-covering. Set $S' = U$, and take as section the S -graph of the inclusion of U in X . $\square$

2. Local determination of a morphism of groups

Let G, H be groups, and let $U \subseteq G$ be a subset such that $U \cdot U = G$. Then, if f, g are homomorphisms from G to H such that $f = g$ on U , one has $f = g$. Similarly:

Proposition 2.1. *Let S be a site (cf. Exp. IV⁽⁵⁾), G a sheaf of groups on S , and $U \subseteq G$ a subsheaf of sets such that the morphism $U \times U \rightarrow G$ induced by multiplication is an epimorphism. Then, if f, g are homomorphisms from G to a sheaf of groups H , equal on U , one has $f = g$.*

Corollary 2.2. *Let G be an S -group prescheme locally of finite presentation and flat, let $U \subseteq G$ be a relatively schematically dense open subset, and H a sheaf of groups on S for the fppf topology. Then every homomorphism $f : G \rightarrow H$ is determined by its restriction to U .*

PROOF. Indeed, since G is flat over S , the composition law in G is a flat morphism (VIB.9.2.xi), and it follows that $U \times_S U \rightarrow G$ is faithfully flat and locally of finite presentation, hence fppf-covering, hence an epimorphism. $\square$

Proposition 2.3. *Let G be a group prescheme locally of finite presentation and flat over S , $U \subseteq G$ a relatively schematically dense open subset, and H a sheaf of groups for the fppf topology. Denote by m_G the multiplication of G , and by m_H that of H .*

Let $\bar{f} : U \rightarrow H$ be an S -morphism, and suppose that there exists a relatively schematically dense open subset V of $m_G^{-1}(U) \cap (U \times_S U)$ such that the diagram

$$\begin{array}{ccc} V & \xrightarrow{(\bar{f} \times \bar{f})|_V} & H \times_S H \\ m_G \downarrow & & \downarrow m_H \\ U & \xrightarrow{\bar{f}} & H \end{array}$$

is commutative. Then there exists a morphism of S -groups $f : G \rightarrow H$ (necessarily unique) extending $\bar{f}$, in each of the following situations:

- (i) H is representable.
- (ii) The diagonal morphism $H \rightarrow H \times_S H$ is represented by a closed immersion. ⁽⁶⁾
- (iii) For every section $a \in U(S)$, the open subset $\text{pr}_2((a \times_S U) \cap V)$ is relatively schematically dense in U , ^(*) and this statement remains true after every base change $S' \rightarrow S$.

PROOF. First note that V is relatively schematically dense in $G \times_S G$. Indeed, V is relatively schematically dense in

$$m_G^{-1}(U) \cap (U \times_S G) \cap (G \times_S U),$$

and the three factors are deduced from $U \subseteq G$ by an evident base change $G \times_S G \rightarrow G$, which implies the assertion by 1.1.

⁽⁵⁾**N.D.E.** That is, a category endowed with a topology, cf. IV, § 4.2.

⁽⁶⁾**N.D.E.** See the note in 1.4.

^(*)Or in G , which amounts to the same by virtue of 1.1 (iv).

To construct a morphism $f : G \rightarrow H$, it suffices, since $U \times_S U \rightarrow G$ is fppf-covering, to find a morphism from $U \times_S U$ to H such that the two induced morphisms on $(U \times_S U) \times_G (U \times_S U)$ are the same. Take $m_H \circ (\bar{f} \times \bar{f}) : U \times_S U \rightarrow H$. One must verify that whenever one has sections $a, b, c, d \in U(S')$, with $S' \rightarrow S$ arbitrary, such that $ab = cd$, one also has $\bar{f}(a)\bar{f}(b) = \bar{f}(c)\bar{f}(d)$. By hypothesis, this is true if (a, d) and (c, d) are contained in $V(S')$, since in this case one has $\bar{f}(a)\bar{f}(b) = \bar{f}(ab)$ and $\bar{f}(c)\bar{f}(d) = \bar{f}(cd)$. Thus it is true whenever $(a, b, c, d) \in (V \times_G V)(S')$.

I say that $V \times_G V$ is an open subset of $(U \times_S U) \times_G (U \times_S U)$ schematically dense relative to S . This will complete the proof in cases (i) and (ii), by 1.4, the facts that the morphism f thus constructed extends $\bar{f}$, and that f is a homomorphism, being evident, also by 1.4. Now write

$$V \times_G V = (V \times_G (G \times_S G)) \cap ((G \times_S G) \times_G V).$$

By symmetry and 1.1, it suffices to verify that $V \times_G (G \times_S G)$ is relatively schematically dense in $(G \times_S G) \times_G (G \times_S G)$. But this last prescheme is S -isomorphic to $G \times_S G \times_S G$, the morphism being given by $(a, b, c, d) \mapsto (a, b, c)$. Thus what one must prove is that $V \times_S G$ is schematically dense in $G \times_S G \times_S G$ relative to S , which is a consequence of the fact that V is relatively schematically dense in $G \times_S G$.

It remains to treat case (iii). To prove $\bar{f}(a)\bar{f}(b) = \bar{f}(c)\bar{f}(d)$, one may make an fppf-covering base extension. Suppose that one has a section $x \in G(S')$, with $S' \rightarrow S$ fppf-covering, such that $(b, x), (a, bx), (d, x), (c, dx)$ are all in V . Then one will have

$$\begin{aligned} \bar{f}(a)\bar{f}(b)\bar{f}(x) &= \bar{f}(a)\bar{f}(bx) = \bar{f}(abx) = \bar{f}(cdx) \\ &= \bar{f}(c)\bar{f}(dx) = \bar{f}(c)\bar{f}(d)\bar{f}(x), \end{aligned}$$

whence the desired equality.

⁽⁷⁾ To find such an x , set, for every $z \in U(S')$,

$$V_z = \text{pr}_2((z \times_{S'} U_{S'}) \cap V_{S'}).$$

Then the hypotheses on x mean that $x \in W$, where

$$W = V_b(S') \cap V_d(S') \cap b^{-1}V_a(S') \cap d^{-1}V_c(S').$$

By (iii), W is relatively schematically dense in G . Thus the existence of a section x after an fppf-covering extension follows from Proposition 1.7.

Let us verify that the morphism thus constructed is multiplicative: let $a, b \in G(S')$, with $S' \rightarrow S$ arbitrary, and write $S' = S$. Choose an fppf-covering base extension $S' \rightarrow S$ and a section $x \in G(S')$ such that $x \in U(S')$ and $ax^{-1} \in U(S')$, that is, $x \in (a^{-1}U)^{-1}(S')$. Choose moreover another fppf-covering extension $S'' \rightarrow S'$ and a section $y \in G(S'')$ such that

$$y, by^{-1}, aby^{-1} \text{ belong to } U(S''),$$

and

$$by^{-1} \in V_x, \quad xby^{-1} \in V_{ax^{-1}}^{-1}.$$

Then, by definition of f , one has over S''

$$\begin{aligned} f(a) &= \bar{f}(ax^{-1})\bar{f}(x), & f(b) &= \bar{f}(by^{-1})\bar{f}(y), \\ f(ab) &= \bar{f}(aby^{-1})\bar{f}(y). \end{aligned}$$

Moreover,

$$\begin{aligned} f(a)f(b) &= \bar{f}(ax^{-1})\bar{f}(x)\bar{f}(by^{-1})\bar{f}(y) \\ &= \bar{f}(ax^{-1})\bar{f}(xby^{-1})\bar{f}(y) \\ &= \bar{f}(aby^{-1})\bar{f}(y) = f(ab), \end{aligned}$$

whence multiplicativity.

⁽⁷⁾**N.D.E.** The beginning of the following sentence has been added.

The fact that f extends $\bar{f}$ is now easy: let $a \in U(S')$, write $S' = S$, and choose $S' \rightarrow S$ fppf-covering and sections $x, y \in G(S')$ such that $(x, y), (ax, y), (a, xy)$ are in $V(S')$. Then

$$\begin{aligned} f(xy) &= \bar{f}(x)\bar{f}(y) = \bar{f}(xy), \\ f(axy) &= \bar{f}(ax)\bar{f}(y) = \bar{f}(axy). \end{aligned}$$

Thus

$$\begin{aligned} f(a) &= f(axy)f((xy)^{-1}) = f(axy)f(xy)^{-1} \\ &= \bar{f}(axy)\bar{f}(xy)^{-1} = \bar{f}(a). \end{aligned}$$

□

Remark 2.3.1. In many cases hypothesis (iii) is true, for it will even be true if one replaces U by a smaller open subset U' , still relatively schematically dense in G . For example, one has:

Proposition 2.4. *With the situation as in 2.3, suppose that each geometric fiber of G/S is irreducible. Let $U' = \text{pr}_1 V$, and*

$$V' = V \cap m_G^{-1}(U') \cap (U' \times_S U').$$

Then $U' \subseteq U$ is a relatively schematically dense open subset of G , and the objects U' , $\bar{f}|_{U'}$, and V' satisfy hypothesis (iii).

PROOF. U' is open, since $G \times_S G \rightarrow G$ is flat and locally of finite presentation. All the other verifications are trivial, except for hypothesis (iii). Let $a \in U(S')$, and write $S' = S$. To verify that $\text{pr}_2((a \times U') \cap V')$ is relatively schematically dense in U' , it suffices to do so fiber by fiber by Corollary 1.3, that is, it suffices to treat the case where S is the spectrum of a field, and in this case it suffices to prove that U' is nonempty, since G is irreducible and “without embedded components” (cf. VIA, 1.1.1). Now

$$\begin{aligned} \text{pr}_2((a \times U') \cap V') &= \text{pr}_2((a \times U') \cap V) \\ &\quad \cap \text{pr}_2((a \times U') \cap m_G^{-1}(U')), \end{aligned}$$

and the second term of the right-hand member is dense in G . Thus it suffices to prove that $\text{pr}_2((a \times U') \cap V)$ is dense in G , that is, nonempty, which is clear since $a \in U' = \text{pr}_1 V$. □

3. Construction of a group from a rational law

3.0. — Suppose that a prescheme X/S and a rational map $X \times_S X \rightarrow X$ over S are given, and seek a group G/S and a map birational relative to S ⁽⁸⁾

$$X \dashrightarrow G$$

which commutes with the composition laws. We treat only the case where X/S satisfies the following hypothesis:

- ($\diamond$) X/S is faithfully flat, of finite presentation,
and has separated fibers “without embedded components”.

(Note that the last two hypotheses are properties true for a group prescheme.⁽⁹⁾)

We shall often suppress the symbol S in fibered products. Let X/S be a prescheme having properties ($\diamond$) above, and let W be a subprescheme of finite presentation of $X \times X \times X$ having the following property:

- (*) The three morphisms $W \rightarrow X \times X$ given by the projections of X^3 onto X^2
are open immersions schematically dense relative to S .

⁽⁸⁾**N.D.E.** Make explicit the notion of birational map (relative to S), perhaps in an addition 1.5.2?

⁽⁹⁾**N.D.E.** Cf. VIA, to be specified...

Notations. We shall use the following terminology: given sections $a, b, c \in X(S')$, with $S' \rightarrow S$ arbitrary, such that $(a, b, c) \in W(S')$, we write:

$$c = ab, \quad b = a^{-1}c, \quad a = cb^{-1}.$$

Definition 3.0.1. ⁽¹⁰⁾ We say, given a section $(a, b) \in X^2(S')$, that ab is defined if and only if there exists a section $c \in X(S')$ such that $(a, b, c) \in W(S')$, i.e. if and only if (a, b) is in $\text{pr}_{12}W(S')$. Similarly, saying that $a^{-1}b$ or ab^{-1} is defined has the analogous meaning, and one also extends this terminology to products of several factors.

Remark 3.0.2. Let us immediately note the following fact: by (i), W defines a rational map $X^2 \rightarrow X$ over S (that given by $(a, b) \mapsto ab$). It may well happen that this rational map has a domain of definition larger than $\text{pr}_{12}W$. Nevertheless, we say that ab is defined only if $(a, b) \in \text{pr}_{12}W(S')$.

Definition 3.1. A *group germ* is a prescheme X/S having properties $(\diamond)$ above, and a subprescheme W of $X \times X \times X$ of finite presentation having the following properties:

- (i) W satisfies property $(*)$ above.
- (ii) For every section $a \in X(S)$, the sets

$$\begin{aligned} \text{pr}_i((a \times X \times X) \cap W), & \quad i = 2, 3, \\ \text{pr}_i((X \times a \times X) \cap W), & \quad i = 3, 1, \\ \text{pr}_i((X \times X \times a) \cap W), & \quad i = 1, 2, \end{aligned}$$

are schematically dense in X relative to S , and this statement remains true after every base change $S' \rightarrow S$. (Hypothesis (i) implies that these sets are open subsets of X .) Intuitively, this means that “ ax is defined for sufficiently general x ”, etc.

(iii) The law is *associative*, that is, if $(a, b, c) \in X^3(S')$, with $S' \rightarrow S$ arbitrary, is such that $(ab)c$ and $a(bc)$ are defined, one has $(ab)c = a(bc)$.

Remarks. (a) In (i), one may replace the condition that W be schematically dense in X^2 relative to S by the condition that, for every $s \in S$, the fiber W_s be dense in X_s^2 in the topological sense, thanks to the hypotheses on X and to 1.2.

(b) Condition (iii) is equivalent to the following: let V_1 (resp. V_2) be the open subset of X^3 where the rational map $(a, b, c) \mapsto a(bc)$ (resp. $(a, b, c) \mapsto (ab)c$) is defined. ^(*) Then there exists an open subset $V \subseteq V_1 \cap V_2$ schematically dense in X^3 relative to S such that the two preceding maps agree on V . This is a consequence of 1.4, since, of course, V_1, V_2 are schematically dense in X^3 relative to S .

(c) Hypothesis (ii) will serve below to ensure that X will be a subobject of the group prescheme which it defines. In many cases, one can deduce (ii) from (i), provided that one replaces X by a relatively schematically dense open subset X' , and W by a relatively schematically dense open subset of $W \cap X'^3$. In fact, one has:

Proposition 3.2. *Suppose that each geometric fiber of X/S is irreducible, and let W be a subprescheme of X^3 satisfying condition (i) of 3.1. Then there exists a relatively schematically dense open subset X' of X , and a relatively schematically dense open subset W' of $W \cap (X' \times X')$, such that the pair (X', W') satisfies conditions (i) and (ii). If (iii) holds for (X, W) , it holds for (X', W') .*

PROOF. Set $X' = \bigcap_{i=1}^3 \text{pr}_i W$. Each $\text{pr}_i W$ is open in X , since $W \rightarrow X^2$ is an open immersion, and the projections $X^2 \rightarrow X$ are flat and of finite presentation. Moreover, $\text{pr}_i W$

⁽¹⁰⁾**N.D.E.** The numbering 3.0.1 and 3.0.2 has been introduced.

^(*)In the sense explained in 3.0.1; one could also replace these open subsets by the domains of definition (cf. 1.5) of the rational maps under consideration.

is relatively schematically dense in X , since W is so in X^2 , and the projections $X^2 \rightarrow X$ are surjective (it suffices to verify density in the topological sense). Take $W' = W \cap X'^3$.

To verify that (i) is true, note that

$$\begin{aligned} W' &= (W \cap (X' \times X' \times X)) \cap (W \cap (X \times X' \times X')) \\ &\quad \cap (W \cap (X' \times X \times X')). \end{aligned}$$

Now $W \cap (X' \times X' \times X) \simeq \text{pr}_{12}W \cap (X' \times X')$, hence is relatively schematically dense in W . Similarly, the other terms of the right-hand member are relatively schematically dense in W , and consequently W' is relatively schematically dense in W . Thus $\text{pr}_{ij}W'$ is relatively schematically dense in $\text{pr}_{ij}W$, hence in $X \times X$, hence in $X' \times X'$.

To verify condition (ii), let $a \in X'(S')$, and write $S' = S$. We must prove that, for example, $\text{pr}_2((a \times X' \times X') \cap W')$ is schematically dense in X' relative to S . By 1.3, it suffices to verify this fiber by fiber, that is, it suffices to treat the case where S is the spectrum of a field, and in this case it suffices to verify that the open subset is nonempty, since the fibers of X/S are irreducible and “without embedded components”. Since $\text{pr}_2((a \times X \times X) \cap W)$ is nonempty, a being a section of X' , this open subset is dense in X . One has

$$\text{pr}_2((a \times X \times X) \cap W) = \text{pr}_2((a \times X) \cap \text{pr}_{12}W).$$

Thus

$$\begin{aligned} \text{pr}_2((a \times X) \cap \text{pr}_{12}W) \cap X' &= \text{pr}_2((a \times X') \cap \text{pr}_{12}W) \\ &= \text{pr}_2((a \times X' \times X) \cap W) \end{aligned}$$

is dense in X , hence in X' . Similarly, $\text{pr}_3(a \times X \times X' \cap W)$ is dense in X , hence in $\text{pr}_3(a \times X \times X \cap W)$; that is, $(a \times X \times X' \cap W)$ is dense in $(a \times X \times X \cap W)$; that is, $\text{pr}_2(a \times X \times X' \cap W)$ is dense in $\text{pr}_2(a \times X \times X \cap W)$, hence dense in X , hence dense in X' . Now, since

$$\begin{aligned} \text{pr}_2(a \times X' \times X' \cap W') &= \text{pr}_2(a \times X' \times X' \cap W) \\ &= \text{pr}_2(a \times X' \times X \cap W) \cap \text{pr}_2(a \times X \times X' \cap W), \end{aligned}$$

it is indeed dense in X' , which is what had to be proved. The other assertions of (ii) follow by symmetry, and the fact that condition (iii) is preserved is trivial. ⁽¹¹⁾ Proposition 3.2 is proved. $\square$

(12)

3.2.1. — Now fix a group germ (X, W) over S . We must make some preliminary remarks on the situation, which we have gathered below. We shall often use these rules without explicit mention in what follows.

Let $a \in X(S')$, and write $S' = S$. Then one obtains a (bi)rational map φ over S from X to itself by associating to a section x the section ax , if it is defined. By 3.1 (ii), the domain of definition of φ contains the relatively schematically dense open subset $\text{pr}_2((a \times X \times X) \cap W)$, and φ defines an isomorphism from this open subset onto the open subset (where $a^{-1}x$ is defined) $\text{pr}_3((a \times X \times X) \cap W)$. This remark is generalized in an evident manner in Rule 1.

Rule 1. Let $P = P(x, t_1, \dots, t_n)$ be a “product” of the symbols $x, t_1, \dots, t_n$, obtained recursively as follows: $P_0 = x$; P_{i+1} is one of the following expressions:

$$P_i t, \quad t P_i, \quad P_i^{-1} t, \quad t P_i^{-1}, \quad P_i t^{-1}, \quad t^{-1} P_i,$$

where t is one of the t_j ; $P_r = P$. Let $a_1, \dots, a_n \in X(S')$. Then there exists a relatively schematically dense open subset U of $X_{S'}$ such that the product $P(x, a_1, \dots, a_n)$ is defined (in the sense of Remark 3.0.2) for a section $x \in X(S'')$ if and only if $x \in U(S'')$, and the map

⁽¹¹⁾**N.D.E.** The following sentence has been added.

⁽¹²⁾**N.D.E.** The numbering 3.2.1, as well as 3.2.2, has been added.

$x \mapsto P(x, (a))$ gives an isomorphism from U onto another relatively schematically dense open subset of $X_{S'}$, denoted $P(U, (a))$.

Rule 2. Let $a, b \in X(S')$. Then:

- If ab is defined, so is $a^{-1}(ab)$, and $a^{-1}(ab) = b$.
- If $a^{-1}b$ is defined, so is $a(a^{-1}b)$, and $a(a^{-1}b) = b$.
- If ba^{-1} is defined, so is $(ba^{-1})a$, and $(ba^{-1})a = b$.

Rule 3. Let $a, b, b' \in X(S')$. If $ab = ab'$, if $ba = b'a$, if $a^{-1}b = a^{-1}b'$, or if $ba^{-1} = b'a^{-1}$, then $b = b'$. Here it is understood that the equality relation implies that both sides are defined.

Rule 4. Let $a, b, c \in X(S')$. Then, whenever both sides are defined, one has

$$a((ba)^{-1}c) = b^{-1}c.$$

Similarly:

$$\begin{aligned} (c(ab)^{-1})a &= cb^{-1}, & a^{-1}((ab^{-1})c) &= b^{-1}c, \\ (c(b^{-1}a))a^{-1} &= cb^{-1}. \end{aligned}$$

Rule 5. All the following associativity laws are true whenever both sides are defined:

$$\begin{aligned} (a^{-1}b)c &= a^{-1}(bc), & (ab^{-1})c &= a(b^{-1}c), & (ab)c^{-1} &= a(bc^{-1}), \\ (ab)^{-1}c &= b^{-1}(a^{-1}c), & (a^{-1}b)c^{-1} &= a^{-1}(bc^{-1}), \\ (ab^{-1})c^{-1} &= a(cb)^{-1}. \end{aligned}$$

3.2.2. — Verification of the rules. (1) is proved by evident induction on the length r of P , the case $r = 1$ being a direct consequence of (3.1) (ii).

(2) This is trivial by definition.

(3) Indeed, by Rule 2, for example in the first case, one has

$$b = a^{-1}(ab) = a^{-1}(ab') = b'.$$

(4) Let us verify, for example, the first relation: on the right-hand side, left multiplication by b is defined and gives c , by Rule 2. Suppose that it is also defined on the left-hand side. Then one will have

$$b(a((ba)^{-1}c)) = (ba)((ba)^{-1}c) = c.$$

Indeed, (ba) is defined by hypothesis, since it occurs in the expression. Thus the middle member is defined and equal to c by Rule 2, and equal to the left-hand member by associativity (cf. 3.1 (iii)). Thus Rule 3 implies that the desired equality is true if this multiplication by b is defined. Now fix a, b . Then Rule 1 implies that $b(a((ba)^{-1}c))$ is indeed defined for c “in” an open subset U of X relatively schematically dense; hence, in this relatively schematically dense open subset, the two rational maps $c \mapsto b^{-1}c$ and $c \mapsto a((ba)^{-1}c)$ are equal. By 1.4, they are equal on every common domain of definition, whence the desired result.

(5) This is the same kind of argument as the preceding one. For example, one verifies $(ab^{-1})c^{-1} = a(cb)^{-1}$ as follows: if right multiplication by c is defined on the right-hand side, one has equality by Rule 4. Since it suffices to verify such a formula on a relatively schematically dense open subset, one is reduced to the case where this multiplication is indeed defined.

3.2.3. — Now consider the relation R on $X \times X$ obtained by setting, for $a, b, a', b' \in X(S')$, $(a, b) \sim (a', b')$ modulo $R(S')$ if and only if there exists an $S' \rightarrow S$ covering for fppf, and a section $x \in X(S')$ such that $(xa)b, (xa')b'$ are defined and equal. Then R is an equivalence relation.

Indeed, this relation is evidently symmetric. By Rule 1, the product $(xa)b$ is defined if x is “in” a suitable relatively schematically dense open subset. Thus 1.7 asserts that there exists

an $S' \rightarrow S$ covering for fppf, and an $x \in X(S')$, such that $(xa)b$ is defined. The relation is therefore reflexive, and transitivity is a consequence of the following lemma:

Lemma 3.3. *Let x, y, a, b, a', b' be sections of $X(S')$ such that $(xa)b, (xa')b', (ya)b, (ya')b'$ are defined. If $(xa)b = (xa')b'$, then $(ya)b = (ya')b'$.*

Indeed, the lemma says that one can test $(a, b) \sim (a', b')$ with an arbitrary $x \in X(S')$, $S' \rightarrow S$ fppf-covering, such that both products are defined. Given $a, b, a', b', a'', b'' \in X(S)$, one may, by Rule 1, 1.1, and 1.7, find an fppf-covering extension $S' \rightarrow S$ and a section $x \in X(S')$ such that the three products in question are defined, whence transitivity.

PROOF OF THE LEMMA. To begin with, write formally:

$$\begin{aligned} (za)b &= (((zx^{-1})x)a)b \\ &= ((zx^{-1})(xa))b = (zx^{-1})((xa)b), \\ (za')b' &= (((zx^{-1})x)a')b' \\ &= ((zx^{-1})(xa'))b' = (zx^{-1})((xa')b'). \end{aligned}$$

One verifies that these equalities are indeed true if the members are defined, by the appropriate rules. ⁽¹³⁾ It follows that $(za)b = (za')b'$ if all these expressions are defined. Moreover, by Rule 1 and the hypotheses already made, these expressions are indeed defined if $z \in X(S'')$ is in $V(S'')$, where V is a certain relatively schematically dense open subset of X (we have taken $S' = S$). Thus the two rational maps from X to itself given by $z \mapsto (za)b$ and $z \mapsto (za')b'$ are equal, whence $(ya)b = (ya')b'$. $\square$

Lemma 3.4. *Consider the rational map $\varphi : X^3 \rightarrow X$ over S , defined by $(a, b, c) \mapsto c^{-1}(ab)$. Let U be the domain of definition of φ , and consider the graph Γ of the morphism $f : U \rightarrow X$ induced by φ , which is a subscheme of X^4 . Then a section $(a, b, c, d) \in X^4(S)$ is in $\Gamma(S)$ if and only if $(a, b) \sim (c, d)$.*

PROOF. First note that the rational map φ is the same as that given by the formula $(a, b, c) \mapsto (xc)^{-1}((xa)b)$ for an arbitrary section $x \in X(S)$. It amounts to the same to say that $c^{-1}(ab) = (xc)^{-1}((xa)b)$ whenever both sides are defined. We leave the verification to the reader.

Let us then show that the map φ is defined for a section $(a, b, c) \in X^3(S)$ if and only if there exists d with $(a, b) \simeq (c, d)$, and that then $d = \varphi(a, b, c)$. Indeed, suppose that one has $(a, b) \sim (c, d)$. To verify that the map φ is defined, one may make an fppf-covering base extension (Proposition 1.6), and may therefore suppose that there exists a section $x \in X(S')$ such that $(xa)b, (xc)d$ are defined and equal. It follows (Rule 2) that $(xc)^{-1}((xa)b)$ is defined and equal to d .

Conversely, suppose that the map φ is defined at the section (a, b, c) , and let $d = \varphi(a, b, c)$. Choose $S' \rightarrow S$ fppf-covering and a section $x \in X(S')$ such that $(xa)b, (xc)d$ are defined. We wish to show that they are equal. For this, it suffices to prove that the two rational maps over S from X to itself, given by $b \mapsto (xa)b$ and $b \mapsto (xc)\varphi(a, b, c)$, are the same, which follows from the remark in the first paragraph. $\square$

Proposition 3.5. *The equivalence relation $R \rightrightarrows X^2$ is representable, and is a flat relation of finite presentation, that is, the projections of R onto X^2 are flat morphisms of finite presentation.*

PROOF. One may suppose S affine. Since (X, W) is of finite presentation over S , one can descend the entire situation to an S of finite type over $\text{Spec } \mathbf{Z}$, hence noetherian. We may therefore suppose that S is noetherian. Then it is trivial that the graph Γ of (3.4) is of

⁽¹³⁾**N.D.E.** To be specified...

finite presentation over X^2 . The projection $\mathrm{pr}_{12}|_\Gamma$ is flat, since $\Gamma \xrightarrow{\sim} \mathrm{pr}_{123}\Gamma$, which is an open subset of X^3 , and the projection $\mathrm{pr}_{12} : X^3 \rightarrow X^2$ is flat because X is flat over S .

I say that Γ represents R . Note that there is something to prove, since the domain of definition of a rational map does not in general commute with base extensions. Let $S'' \rightarrow S$. What is clear, by (3.4) applied to S'' , is that $R(S'') \supset \Gamma(S'')$, since $\varphi \times_S S''$ is certainly defined on $U \times_S S''$. Thus let $(a, b, c, d) \in R(S'')$. One must prove that $(a, b, c, d) \in \Gamma(S'')$. The verification is made locally for, say, the étale topology. One may therefore suppose that S'' is strictly local, i.e. the spectrum of a henselian ring with separably closed residue field. Moreover, applying (1.6) and the customary sorites of passage to the limit, one is reduced to the case where S is strictly local and $S'' \rightarrow S$ is local. Suppose that one has a section $x \in X(S)$ such that over S'' the products $(xa)b, (xc)d$ are defined. This will imply that $(xc)^{-1}((xa)b)$ is defined and equal to d . Now there exists a relatively schematically dense open subset V of X^3 such that $(xc)^{-1}((xa)b)$ is defined if and only if $(a, b, c) \in V(S'')$, and one has $V \subseteq U$. Thus $(a, b, c, d) \in \Gamma(S'')$ if such an x exists. By (1.6), one may make an fppf-covering base extension $S' \rightarrow S$ to find such an x . Since S is strictly local, one can (*) find $S' \rightarrow S$ faithfully flat, local, and finite, and a section $x \in X(S')$ which “passes through” an arbitrary closed point of the closed fiber of X/S . Now saying that $(xa)b$ and $(xc)d$ are defined means that, over S'' , x is in a certain relatively schematically dense open subset, which is checked on the closed fiber of $X_{S''}$. Thus it works. $\square$

Now let G be the quotient of X^2 by R as a sheaf for the fppf topology. We shall define a composition law on G as follows: let $(g, g') \in G(S')$ be represented by a section $((a, b), (c, d))$ of $X^2 \times X^2(S'')$, with $S'' \rightarrow S'$ fppf-covering. Suppose moreover that X admits a section x over S'' such that $a(b(cx))$ and $x^{-1}d$ are defined, which is permitted by Rule 1 and (1.7), and call gg' the class in $G(S')$ represented by the section $(a(b(cx)), x^{-1}d)$ of $X^2(S'')$.

Let us verify that gg' does not depend on the choice of the section x and the representative $((a, b), (c, d))$: indeed, let $(a', b') \sim (a, b)$, $(c', d') \sim (c, d)$, and x be such that $a'(b'(c'x'))$ and $x'^{-1}d'$ are defined. We may suppose that all are sections over S'' . One must prove that

$$(a(b(cx)), x^{-1}d) \sim (a'(b'(c'x')), x'^{-1}d'),$$

that is, for a suitable section $z \in X(S''')$, with $S''' \rightarrow S''$ suitably fppf-covering, one has

$$(z(a(b(cx))))(x^{-1}d) = (z(a'(b'(c'x'))))(x'^{-1}d').$$

Whenever all the products are defined, one has

$$\begin{aligned} (z(a(b(cx))))(x^{-1}d) &= ((za)(b(cx)))(x^{-1}d) \\ &= (((za)b)(cx))(x^{-1}d) \\ &= (((za)b)c)x)(x^{-1}d) \\ &= (((za)b)c)(x(x^{-1}d)) = (((za)b)c)d, \end{aligned}$$

and the same identities hold with the primes. Now, by Rule 1 and (1.7), such a z exists. One must therefore prove that

$$(((za)b)c)d = (((za')b')c')d'.$$

But $(za)b = (za')b'$, since $(a, b) \sim (a', b')$ (3.3), and one therefore has the desired equality, since $(c, d) \sim (c', d')$.

Consider the natural morphism $i : X \rightarrow G$ defined as follows: for $a \in X(S')$, choose $S'' \rightarrow S'$ fppf-covering and a section $b \in X(S'')$ such that ab^{-1} is defined, and set

$$i(a) = \text{class in } G \text{ of } (ab^{-1}, b).$$

One easily verifies that this class, which is a priori in $G(S'')$, does not depend on the choice of b , and thus gives a well-defined element of $G(S')$. The reader will take pleasure in verifying the

(*)Combining Exp. VIA, 1.1.1 and EGA IV₄, 17.16.2.

Proposition 3.6. *The morphism i commutes with the composition laws of X and G , that is, if $a, b \in X(S')$ are sections such that ab is defined, one has $i(a)i(b) = i(ab)$.*

The aim of this number is the following theorem:

Theorem 3.7. ⁽¹⁴⁾ *Let (X, W) be a group germ over S , with X/S faithfully flat, of finite presentation, and with separated fibers without embedded components. Then, with the notation above, one has*

- (i) G is a sheaf of groups.
- (ii) $i : X \rightarrow G$ is represented by an open immersion.
- (iii) G is representable locally on S for the fppf topology.
- (iv) If G/S is representable, then it is a flat group of finite presentation, and $i : X \rightarrow G$ is schematically dense relative to S .

Note that G is evidently characterized by properties (i), ..., (iv); one may therefore forget the explicit construction of G . The proof proceeds in several steps:

Lemma 3.8. (i) *Let c be a section of $X(S)$; then the morphism from the prescheme $c \times X$ (which is S -isomorphic to X) to G , given by $c \times X \hookrightarrow X \times X \rightarrow G$, is a monomorphism.*

(ii) *Let $\{c_i\}$, $i \in I$, be sections of $X(S)$, let $Z = \coprod_i c_i \times X$, and call $R' \rightrightarrows Z$ the equivalence relation induced on Z by the evident morphism $Z \rightarrow X^2$. Then R' is a "gluing" of the $c_i \times X \simeq X$.*

PROOF. (i) For two sections $(c, a), (c, a')$ of $(c \times X)(S')$ to have the same image in G , one must have $(c, a) \sim (c, a')$, that is, $(xc)a \sim (xc)a'$ for a suitable $x \in X(S'')$, with $S'' \rightarrow S'$ fppf-covering, whence $a = a'$ by Rule 3.

(ii) Let c_i, c_j be two sections, and consider the birational map ψ_{ji} from X to itself over S , given by the formula $x \mapsto c_j^{-1}(c_i x)$. This is the same map as that given by $x \mapsto (yc_j)^{-1}((yc_i)x)$, if $y \in X(S)$, as one easily sees. Moreover, one verifies that φ_{ji} is defined for a $b \in X(S')$ if and only if there exists b' such that $(c_i, b) \sim (c_j, b')$, and then $b' = \varphi_{ji}(b)$. Let U_{ji} be the domain of definition over S of φ_{ji} . It remains to prove that this domain of definition is universal, that is, if $b \in X(S'')$, with $S'' \rightarrow S$ arbitrary, and φ_{ji} is defined at b , then $b \in U_{ji}(S'')$. It amounts to the same to prove that if $b, b' \in X(S'')$ are such that $(c_i, b) \simeq (c_j, b')$, then $b \in U_{ji}(S'')$. We leave the verification of this fact, analogous to that of (3.5), to the reader. $\square$

Lemma 3.9. *Suppose that $\{c_i\}$, $i \in I$, are sections of $X(S)$ such that $\coprod_i c_i \times X \rightarrow G$ is surjective as a morphism of sheaves. Then G is representable and flat, of finite presentation over S , and the structural morphism $X^2 \rightarrow G$ is flat and of finite presentation.*

PROOF. The fact that G is representable is an immediate consequence of (3.8), and it follows that $\coprod_i c_i \times X \rightarrow G$ is an open covering. To prove that $X^2 \rightarrow G$ is flat, it suffices to do so locally, hence to prove that the rational map $X^2 \rightarrow c_i \times X$ induces a flat morphism on its domain of definition. Now this rational map is given by $(a, b) \mapsto (c_i, ((xc_i)^{-1}(xa))b)$, with x an arbitrary section, and, if it is defined at $(a, b) \in X^2(S')$, one can find $S'' \rightarrow S'$ fppf-covering and a section $x \in X(S'')$ such that $((xc_i)^{-1}(xa))b$ is defined. One easily sees that it is therefore a flat morphism. It follows similarly that it is locally of finite presentation, hence fppf-covering. Now, by construction, the relation R is effective. Thus, by (3.5), the morphism $X^2 \rightarrow G$ becomes of finite presentation after the base change $G \leftarrow X^2$, which is fppf-covering; hence $X^2 \rightarrow G$ is of finite presentation. Let us show that $G \rightarrow S$ is flat and of finite presentation. It is flat and locally of finite presentation, since G is covered by the $c_i \times X \simeq X$. Now X^2/S is quasi-compact, and $X^2 \rightarrow G$ is surjective. This proves that G/S is quasi-compact. To prove that $G \rightarrow S$ is quasi-separated, note that one has the following

⁽¹⁴⁾**N.D.E.** If X is smooth, separated over S , faithfully flat of finite presentation over S , then G/S is represented by a smooth S -group scheme of finite type over S . This is Theorem 6.6.1 of the book *Néron models* by Bosch–Lütkebohmert–Raynaud, Springer (1990).

cartesian diagram:

$$\begin{array}{ccc} R & \xrightarrow{\alpha} & X^2 \times_S X^2 \\ \gamma \downarrow & & \downarrow \beta \\ G & \xrightarrow[\Delta]{} & G \times_S G \end{array}$$

One has γ surjective and α, β quasi-compact, hence $\beta\alpha$ is quasi-compact, hence Δ is quasi-compact. ⁽¹⁵⁾ $\square$

Lemma 3.10. *Let $\{c_i\}$, $i \in I$, be sections of $X(S)$. For $\coprod_i c_i \times X \rightarrow G$ to be surjective as a morphism of fppf-sheaves, it suffices that the following condition hold:*

For every $S' \rightarrow S$ and $(a, b) \in X^2(S')$, there exists an open covering $\{S'_\nu\}$, $\nu \in N$, of S' , and a function $N \rightarrow I$, $\nu \mapsto i(\nu)$, such that $(c_{i(\nu)}^{-1}a)b$ is defined over S'_ν .

PROOF. Let $S'' \rightarrow S$ be arbitrary, and $g \in G(S'')$. Choose $S' \rightarrow S''$ fppf-covering, and a section $(a, b) \in X^2(S')$ representing g . Take the open covering $\{S'_\nu\}$ of S' which exists by the hypothesis of the lemma. Then, over each S'_ν , one has $(a, b) \sim (c_{i(\nu)}, (c_{i(\nu)}^{-1}a)b)$; hence g is represented by a section of $[c_{i(\nu)} \times X](S')$ over S'_ν , which proves surjectivity, since the family of morphisms $\{S'_\nu \rightarrow S''\}$ is fppf-covering. $\square$

Lemma 3.11. *Let Y/S be a prescheme of finite presentation, and let $\{a_i\}$, $i \in I$, be sections of $Y(S)$. Let s_0, s_1 be points of S such that s_0 is a specialization of s_1 , and Y_j the fiber of Y/S at s_j . Let C_j be the closure in Y_j of the set of points $\{a_i(s) \cap Y_j\}$. Then one has $\dim C_1 \geq \dim C_0$.*

PROOF. It suffices to make the verification after a base change $S' \rightarrow S$ with chosen points s'_0, s'_1 such that $s'_j \mapsto s_j$ and s'_0 is a specialization of s'_1 . One is therefore (EGA II 7.1.4) reduced to the case where S is the spectrum of a valuation ring A , s_0 the closed point of S , and s_1 the generic point of S . Now let V be the closure of C_1 in Y . It is clear that $C_0 \subseteq V$, and thus the lemma is a consequence of the “well-known” fact that an irreducible closed subscheme V of a prescheme Y/S of finite presentation satisfies $\dim(V \times s_1) \geq \dim(V \times s_0)$ if S is the spectrum of a valuation ring and $V \times s_1 \neq \emptyset$ (EGA IV 13.1.6). $\square$

Lemma 3.12. *Suppose that S is the spectrum of a local ring, with closed point s_0 , and let $\{c_i\}$, $i \in I$, be sections such that the closure C_0 of the set $\{c_i(s) \cap X_0\}$ in the closed fiber X_0 has dimension equal to $\dim X_0 = n$. Then the condition of Lemma 3.9 is satisfied.*

PROOF. First note that the fibers of X/S all have the same dimension u , which follows from EGA IV 12.1.1 (i), and the fact that X has a rational composition law. Lemma (3.11) therefore implies that, for every morphism $S_1 \rightarrow S$ with S_1 the spectrum of a field, the dimension of the closure of the set $\{c_i \times_S S_1\}$ in X_{S_1} is equal to n . Let us verify the condition of (3.10): let $(a, b) \in X^2(S')$. For $(c_i^{-1}a)b$ to be defined, it is necessary and sufficient that c_i be contained in a certain open subset $U \subseteq X_{S'}$ schematically dense relative to S' (Rule 1). One must prove that this is true for a suitable i , locally on S' . It therefore suffices to treat the case where S' is the spectrum of a local ring; then the fact that $c_i \in U(S')$ is checked on the closed fiber. One is therefore reduced to the case $S' = \text{Spec } k$, with k a field. Now, with the above notation, take $S' = S_1$. One has $\dim C_1 = \dim X_{S_1}$, and U is relatively schematically dense in X_{S_1} . Thus $U \cap C_1 \neq \emptyset$, whence $U \cap \{c_p \times_S S_1\} \neq \emptyset$, and one has won. $\square$

The proof of the theorem is now easy. First note the following consequence of the finiteness assertion of Lemma (3.9): if $\{A_i\}$ is an inductive system of rings over S , if $\underline{A} = \varinjlim A_i$, and if the hypotheses of (3.9) hold for $S = \text{Spec } \underline{A}$, then one can descend the object representing the

⁽¹⁵⁾**N.D.E.** Another way to conclude here is by faithfully flat descent (EGA IV₂, 2.7.1), since β is covering for fppf.

quotient G of $R \rightrightarrows X^2$ to one of the $S_i = \operatorname{Spec} A_i$, with the finiteness and flatness properties stated in (3.9). This is the customary passage to the limit (EGA IV 8 and 11). It follows that, for the proof of (iii) and (iv) of (3.7), one may restrict oneself to the case $S = \operatorname{Spec} A$, with A a strictly local ring. Then let x_0 be a closed point of the closed fiber X_0 of X/S . There exists ^(*) an extension A' of A , local, free and finite, and a section of $X' = X \times_S S'$ passing through the unique point x'_0 of X'_0 over x_0 . Note that $X'_0 \rightarrow X_0$ is radicial, since S is strictly local, hence the residue field of A is separably closed. It follows that there exists an inductive system $\{A_i\}$ of local rings, flat and finite over S , such that, setting

$$\underline{A} = \varinjlim A_i, \quad \underline{X} = X \times_S \operatorname{Spec} \underline{A},$$

$\underline{X}$ has a set of sections inducing a dense set on the closed fiber $\underline{X}_0$. For every closed point x_0 of X_0 , take an extension $A(x_0)$ such that the corresponding base extension of X admits a section “passing through x_0 ”, and take as inductive system the system of finite tensor products of the $A(x_0)$. Note that $\underline{A}$ is local, being a limit of local rings. Thus, by (3.12) and (3.10), one has the quotient G over $\operatorname{Spec} \underline{A}$, hence over one of the $\operatorname{Spec} A_i$ by the remarks above, hence locally for the fppf topology.

In fact, it follows from the constructions that locally for the fppf topology one can find a finite set of sections $\{c_i\}$, $i = 1, \dots, r$, such that G is covered by the $c_i \times X$. (ii) and (iv) follow easily from this fact, and we leave the verification of (i) to the reader.

Corollary 3.13. *The sheaf of groups G determined by a group germ (X, W) over S is representable in each of the following situations:*

- (i) S is artinian.
- (ii) For every local scheme $S' \rightarrow S$ of a closed point s of S , $X_{S'}$ has a set of sections inducing on the closed fiber a set whose closure has dimension $\dim X_s$.
- (iii) S is strictly local, and X/S is smooth.
- (iv) There exists $S' \rightarrow S$ fppf-covering such that $G_{S'}$ is representable and affine over S .

PROOF. Indeed, (ii) is a consequence of (3.10) and (3.12); (iii) follows directly from (ii) and “Hensel’s lemma”; (iv) from descent of affine schemes; and (i) from descent of group schemes, which is possible here since one knows that every finite subset of a group over a field is contained in an affine open subset (Exp. VI). $\square$

^(*)Cf. the note at the bottom of page 15, Exp. VII.

Bibliography

- [1] A. Weil, *Variétés abéliennes et courbes algébriques*. Hermann, Paris, 1948.
- [2] H. Yanagihara, *Reduction of group varieties and transformation spaces*. Journ. Sci. Hiroshima Univ., Ser. A-I, vol. 27, no. 1, June 1963.